

Fiolina Germi

Sprite Sheet Documentation

Pistol Configuration — Complete Reference

Character: Fiolina Germi (Female Soldier)

Weapon: Pistol Only

Sprite Sheet File: Fiolina Germi.png

Location: /home/ahmad-ali/Downloads/FInal Backup 7_30/Sprites/

Coordinate System: x = left offset, y = top offset, w = width, h = height (pixels)

Note

All coordinates reference the full **Fiolina Germi.png** sprite sheet. Each entry gives the top-left corner (**x**, **y**) and dimensions **w** × **h** of the bounding rectangle for that frame. The document is ordered to match the provided image sequence (Image 1 through Image 17).

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1 1

1.0 1

This animation plays whenever the character is airborne and descending *without* a parachute. Frames progress from a fully extended fall posture (tall) to a compact, landing-ready crouch (short). The shrinking height across frames gives the visual illusion of the character approaching the ground.



Figure 1: Falling from air — full frame strip (Image 1)

Frame	x	y	w	h
Frame 1	10	18	29	234
Frame 2	44	42	29	210
Frame 3	78	113	29	139
Frame 4	112	139	29	113
Frame 5	146	166	29	86
Frame 6	180	195	29	57
Frame 7	214	217	23	35

i Note

Notice that **y increases** and **h decreases** each frame — this is because the character is positioned lower on the sheet as it approaches the ground, and the visible body shrinks into a compressed landing pose.

2 1

2.0 1

The default idle loop played whenever the character is stationary and not performing any action. Two sub-layers are provided: **Top** (upper body) and **Bottom** (lower body). They are composited at the hip anchor point.

2.1 1



Figure 2: Standing idle strip (bottom area of Image 1)

Frame	x	y	w	h
Frame 1	10	257	28	26
Frame 2	43	257	28	26
Frame 3	76	257	28	26
Frame 4	109	257	28	26

2.2 1

Frame	x	y	w	h
Frame 1	142	259	20	24

 Rule

Loop frames 1–4 of the top body. The single bottom-body frame is held static for the entire idle loop. Composite: `upper_body_bottom_y + hip_offset = lower_body_top_y`.

3 1

3.0 1

Fast movement loop — triggers when the player holds the run button and moves horizontally. Two sub-layers: **Top Body** and **Bottom Body**. Both must stay in frame-sync (frame 1 top + frame 1 bottom, etc.).



Figure 3: Running — top body strip (Image 2)



Figure 4: Running — bottom body strip (Image 3)

3.1 1

Frame	x	y	w	h
Frame 1	10	500	27	26
Frame 2	42	501	27	26
Frame 3	74	502	27	26
Frame 4	106	501	27	26
Frame 5	138	500	27	26
Frame 6	170	500	28	26
Frame 7	203	501	28	26
Frame 8	236	502	28	26
Frame 9	269	502	29	26
Frame 10	303	503	28	26
Frame 11	336	501	26	27
Frame 12	367	501	27	27

3.2 1

Frame	x	y	w	h
Frame 1	10	534	22	26
Frame 2	37	534	25	25
Frame 3	67	535	27	23
Frame 4	99	534	25	26
Frame 5	129	534	17	26
Frame 6	151	535	18	25
Frame 7	174	535	21	25
Frame 8	200	535	26	23
Frame 9	231	535	30	25
Frame 10	266	535	24	25
Frame 11	295	534	18	26
Frame 12	318	534	20	26

🎮 Rule

12-frame loop. Top and bottom bodies share the same frame index. Because this is the **Universal Lower Body**, these bottom frames are reused for all weapon configurations (only the top body swaps on weapon change).

4 1

4.0 1

Plays during slow / cautious movement — typically when the player taps move without holding run, or during story segments. Uses a 4-frame loop.



Figure 5: Slow walk strip (Image 4)

Frame	x	y	w	h
Frame 1	10	657	29	34
Frame 2	44	658	29	33
Frame 3	78	658	29	33
Frame 4	112	657	30	34

i Note

The slow walk lower body (universal) is shared from Section 3 bottom frames. Only the top body is listed here; use the running bottom body (Section 3.2) unless a dedicated slow-walk lower body is specified.

5 1

5.0 1

Triggered when the character picks up a food item or consumes a power-up during gameplay. This is a long 52-frame sequence covering the full eat gesture — bending, consuming, and returning to normal posture.

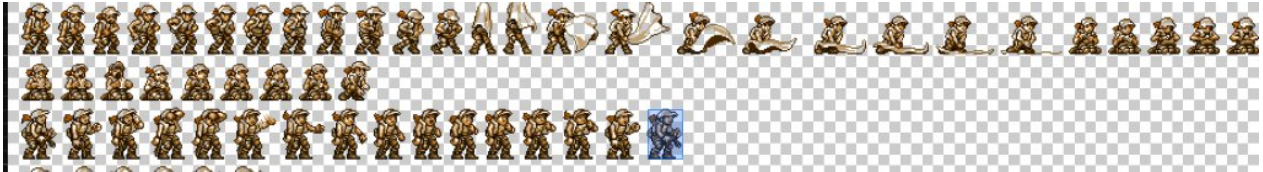


Figure 6: Eat / item pickup animation strip (Image 5)

5.1 1

Frame	x	y	w	h
Frame 1	10	338	21	37
Frame 2	36	339	21	36
Frame 3	62	340	21	35
Frame 4	88	339	24	36
Frame 5	117	339	23	36
Frame 6	145	339	23	36
Frame 7	173	339	22	36
Frame 8	200	339	22	36
Frame 9	227	339	23	36
Frame 10	255	340	23	35
Frame 11	283	342	23	33
Frame 12	311	342	23	33
Frame 13	339	336	20	39
Frame 14	364	333	25	42
Frame 15	394	341	40	34
Frame 16	439	335	48	40
Frame 17	492	344	43	31
Frame 18	540	346	40	29
Frame 19	593	347	38	28
Frame 20	636	347	41	28
Frame 21	682	347	43	28
Frame 22	730	347	45	28
Frame 23	780	348	24	27
Frame 24	809	349	24	26
Frame 25	838	348	24	27
Frame 26	867	348	24	27
Frame 27	896	348	24	27

5.2 1

Frame	x	y	w	h
Frame 28	10	382	24	27
Frame 29	39	382	24	27
Frame 30	68	380	24	29
Frame 31	97	383	25	26
Frame 32	127	382	24	27
Frame 33	156	382	24	27
Frame 34	185	382	24	27
Frame 35	214	382	24	27
Frame 36	243	380	24	29

5.3 1

Frame	x	y	w	h
Frame 37	10	415	26	37
Frame 38	41	415	28	37
Frame 39	74	415	26	37
Frame 40	105	414	25	38
Frame 41	135	415	26	37
Frame 42	166	415	31	37
Frame 43	202	415	31	37
Frame 44	238	415	26	37
Frame 45	269	415	23	37
Frame 46	297	415	22	37
Frame 47	324	415	22	37
Frame 48	351	415	22	37
Frame 49	378	415	25	37
Frame 50	408	415	26	37
Frame 51	439	415	26	37
Frame 52	470	415	26	37

ⓘ Note

This is a **one-shot** animation (plays once, does not loop). After frame 52 the character returns to the standing idle (Section 2). Frames 13–22 contain the wide eating-arc posture — note the significantly larger **w** values (up to 48 px) during the actual consumption gesture.

6 1

6.0 1

The character whistles or signals — used in cut-scenes, scripted events, or when calling for support. A 16-frame one-shot sequence covering the raise, whistle, and lower gesture.



Figure 7: Neat call / whistle animation (Image 6)

Frame	x	y	w	h
Frame 1	10	292	28	36
Frame 2	43	292	27	36
Frame 3	75	292	24	36
Frame 4	104	291	25	37
Frame 5	134	291	25	37
Frame 6	164	289	25	39
Frame 7	194	290	24	38
Frame 8	223	289	24	39
Frame 9	252	288	24	40
Frame 10	281	293	31	35
Frame 11	317	293	33	35
Frame 12	355	295	34	33
Frame 13	394	295	28	33
Frame 14	427	295	25	33
Frame 15	457	294	23	34
Frame 16	485	293	22	35

7 1

7.0 1

Two jump types are defined — each split into top and bottom body layers. Uses the **Palindrome / Ping-Pong** technique: play frames forward on ascent, reverse on descent. This eliminates a separate falling animation.



Figure 8: Jump animations — both types (Image 10)

7.1 1

Frame	x	y	w	h
Frame 1	10	821	29	26
Frame 2	44	823	30	24
Frame 3	79	825	30	22
Frame 4	114	825	30	22
Frame 5	149	825	29	22
Frame 6	183	825	28	23

7.2 1

Frame	x	y	w	h
Frame 1	10	853	22	30
Frame 2	37	853	22	30
Frame 3	64	853	22	29
Frame 4	91	853	23	28
Frame 5	119	853	24	25
Frame 6	148	853	24	22

7.3 1

Frame	x	y	w	h
Frame 1	10	924	24	31
Frame 2	39	924	28	32
Frame 3	72	924	32	31
Frame 4	109	924	28	34
Frame 5	142	924	23	31
Frame 6	170	924	23	28

7.4 1

Frame	x	y	w	h
Frame 1	10	963	28	23
Frame 2	43	963	29	21
Frame 3	78	963	33	19
Frame 4	116	963	30	17
Frame 5	151	963	30	17
Frame 6	186	963	26	19

🎮 Rule

Palindrome rule: Play frames 1→6 on ascent ($\text{velocity_y} < 0$), then 6→1 on descent ($\text{velocity_y} > 0$). Top and bottom bodies stay in sync.

8 1

8.0 1

Three crouch states: the transition *into* a crouch, the static crouched idle, and the crouched walk cycle.



Figure 9: Crouching animation set (Image 7)

8.1 1

Frame	x	y	w	h
Frame 1	10	1778	29	38
Frame 2	44	1776	28	40
Frame 3	77	1777	35	39

8.2 1

Frame	x	y	w	h
Frame 1	10	1821	35	25
Frame 2	50	1821	35	25
Frame 3	90	1821	35	25
Frame 4	130	1821	35	25

8.3 1

Frame	x	y	w	h
Frame 1	10	1851	34	25
Frame 2	49	1851	32	25
Frame 3	86	1851	32	25
Frame 4	123	1851	31	25
Frame 5	159	1851	35	25
Frame 6	199	1851	34	25
Frame 7	238	1851	33	25

ⓘ Note

When crouching, the lower body sprite is attached *directly below* the crouching upper body. The crouch has its own dedicated lower body — do not substitute the running lower body

here.

9 1

9.0 1

All pistol fire sequences follow the same structural rule:

 **CRITICAL**

Pistol firing rule: USE ALL FRAMES — do *not* loop only the first 3 frames for auto-fire (that rule applies to the machine gun). The first 6 frames capture the muzzle-flash / recoil burst. Frames 7 onward are the recovery / return to aimed posture. For continuous fire: repeat frames 1–6. For a single shot: play all frames once.

9.1 1



Figure 10: Straight fire (standing) and straight fire (jumping) — Image 8. Note: label *USED WHEN JUMPING FORWARD* marks the lower row.

 **Note**

This is the **top body only**. It is used during movement as well as standing. Combine with the appropriate lower-body layer (running, walking, or standing idle) at the hip anchor.

Frame	x	y	w	h
Frame 1	10	994	48	19
Frame 2	63	994	33	19
Frame 3	119	993	31	20
Frame 4	175	992	29	21
Frame 5	209	991	29	22
Frame 6	243	994	33	19
Frame 7	281	994	33	19
Frame 8	319	994	33	20
Frame 9	357	994	32	23
Frame 10	394	993	30	25

9.2 1

Frame	x	y	w	h
Frame 1	10	1026	48	19
Frame 2	63	1026	33	19
Frame 3	118	1025	31	20
Frame 4	174	1024	29	21
Frame 5	208	1023	29	22
Frame 6	242	1026	33	19
Frame 7	280	1026	33	19
Frame 8	318	1026	33	20
Frame 9	356	1026	32	23
Frame 10	393	1025	30	25

9.3 1



Figure 11: Straight fire while crouching (Image 9)

Frame	x	y	w	h
Frame 1	10	1914	51	26
Frame 2	66	1914	44	26
Frame 3	123	1913	32	27
Frame 4	180	1912	30	28
Frame 5	215	1911	30	29
Frame 6	250	1914	35	26
Frame 7	290	1914	35	26
Frame 8	330	1914	35	26
Frame 9	370	1914	34	26
Frame 10	409	1913	35	27

ⓘ Note

Frames 1–6 = active firing loop. Frames 7–10 = recovery (transition back to crouched idle or crouch walk).

9.4 1



Figure 12: Top fire strip (Image 11)



Figure 13: Top fire auxiliary frames (Image 12)

Frame	x	y	w	h
Frame 1	10	1654	19	59
Frame 2	34	1677	20	36
Frame 3	59	1679	19	34
Frame 4	83	1680	19	33
Frame 5	107	1681	19	32
Frame 6	131	1680	18	33
Frame 7	154	1682	20	31
Frame 8	179	1683	23	30
Frame 9	207	1687	22	26
Frame 10	234	1690	24	23

9.5 1

Frame	x	y	w	h
Frame 1	10	2441	18	54
Frame 2	33	2441	18	55
Frame 3	56	2440	18	56
Frame 4	79	2441	22	30
Frame 5	106	2441	18	35
Frame 6	129	2441	18	36

9.6 1



Figure 14: Sliding and sliding fire strip (Image 13)

Frame	x	y	w	h
Frame 1	10	2299	34	22
Frame 2	49	2296	31	25
Frame 3	85	2297	31	24
Frame 4	121	2298	35	23
Frame 5	161	2300	46	21

9.7 1

Frame	x	y	w	h
Frame 1	10	2329	50	21
Frame 2	65	2328	46	21
Frame 3	120	2327	46	22
Frame 4	175	2327	46	23
Frame 5	226	2326	46	24

10 1

10.0 1

Triggered when the magazine is empty or the player manually reloads. A long 21-frame sequence covering the full reload gesture — lowering the gun, swapping the clip, and raising back to aimed posture.



Figure 15: Full reload animation strip (Image 14)

Frame	x	y	w	h
Frame 1	10	1064	32	23
Frame 2	47	1059	29	28
Frame 3	81	1057	28	32
Frame 4	114	1063	38	25
Frame 5	157	1063	38	24
Frame 6	200	1063	39	22
Frame 7	244	1063	39	27
Frame 8	288	1063	34	25
Frame 9	327	1063	34	25
Frame 10	366	1063	35	24
Frame 11	406	1063	33	25
Frame 12	444	1063	31	25
Frame 13	480	1063	31	25
Frame 14	516	1063	35	24
Frame 15	556	1063	35	25
Frame 16	596	1063	47	23
Frame 17	648	1056	44	33
Frame 18	704	1055	47	39
Frame 19	756	1063	52	35
Frame 20	813	1063	54	36
Frame 21	872	1063	54	36

 Note

One-shot animation — plays once, does not loop. Frames 17–21 are notably wider (44–54 px) because the character holds the magazine/clip away from the body during the swap phase.

11 1

11.0 1

Close-range melee attack. Two configurations: **standing melee** and **crouching melee**. Both are top-body animations composited on the appropriate lower body.

11.1 1



Figure 16: Milli attack (standing) strip (Image 15)

Frame	x	y	w	h
Frame 1	10	1150	22	30
Frame 2	37	1150	26	34
Frame 3	68	1149	35	32
Frame 4	116	1147	46	34
Frame 5	167	1139	43	40
Frame 6	215	1138	42	41

11.2 1



Figure 17: Milli attack (crouching) strip (Image 16)

Frame	x	y	w	h
Frame 1	10	2095	36	25
Frame 2	51	2095	35	25
Frame 3	91	2095	34	25
Frame 4	130	2091	35	29
Frame 5	178	2089	46	31
Frame 6	229	2081	43	39
Frame 7	277	2080	42	40

12 1

12.0 1

Plays when the player switches from the pistol to another weapon (or vice versa). A 10-frame sequence covering the holster-draw gesture. After frame 10 the new weapon's idle / aim frame takes over.



Figure 18: Gun switching strip (Image 17)

Frame	x	y	w	h
Frame 1	10	1480	25	24
Frame 2	40	1480	26	27
Frame 3	71	1480	28	25
Frame 4	104	1481	29	24
Frame 5	138	1481	30	23
Frame 6	173	1480	28	32
Frame 7	206	1480	26	31
Frame 8	237	1480	28	26
Frame 9	270	1481	34	19
Frame 10	309	1479	32	21

i Note

One-shot transition animation. The lower body remains in standing idle throughout. After frame 10 immediately snap to the first frame of the new weapon's aim pose.

13 1

Animation	Frames	Section	Type	Notes
Falling (no chute)	7	1	One-shot	Height shrinks per frame
Standing Idle — Top	4	2.1	Loop	
Standing Idle — Bot	1	2.2	Static hold	
Running — Top	12	3.1	Loop	Universal top
Running — Bottom	12	3.2	Loop	Universal lower
Slow Walk	4	4	Loop	Uses running lower
Eat / Pickup	52	5	One-shot	3-row layout
Neat Call	16	6	One-shot	
Straight Jump Top	6	7.1	Palindrome	
Straight Jump Bot	6	7.2	Palindrome	
Running Jump Top	6	7.3	Palindrome	
Running Jump Bot	6	7.4	Palindrome	
Crouch Transition	3	8.1	One-shot	
Crouch Idle	4	8.2	Loop	
Crouch Walk	7	8.3	Loop	
Straight Fire	10	9.1	Top only	Frames 1–6 fire
Fire While Jumping	10	9.2	Top only	Frames 1–6 fire
Fire While Crouching	10	9.3	Full body	Frames 1–6 fire
Top Fire	10	9.4	Top only	
Bottom Fire	6	9.5	Top only	
Slide (no fire)	5	9.6	One-shot	
Sliding Fire	5	9.7	One-shot	
Reload	21	10	One-shot	Wide at frames 17-21
Milli — Standing	6	11.1	One-shot	Top only
Milli — Crouching	7	11.2	One-shot	Top only
Gun Switch	10	12	One-shot	

14 1

The following rules govern how all animations in this document are combined, sequenced, and rendered in a production game engine. Techniques are drawn from *Metal Slug*, *Cuphead*, *Guilty Gear*, *Street Fighter*, and similar sprite-based titles.

14.1 1

Technique: Composite Layering

(*Metal Slug*, *Cuphead*, *Guilty Gear*)

Attach the upper body to the lower body at the **hip joint anchor point**. Keep a consistent Y offset between the base of the upper sprite and the top of the lower sprite. This offset must remain constant across all frames for a seamless seam.

$$\text{upper_body_ottom}_y + \text{hip_offset} = \text{lower_body_top}_y$$

🛠 Rule

- For **crouching**: lower body is always attached directly below the crouching upper body.
- For **standing** / **running**: use the universal lower body (Sections 3.2 and 4).
- The hip offset must be validated per animation set to avoid vertical "pop".

14.2 1

Technique: Palindrome Animation

(*Street Fighter*, *Metal Slug series*)

Used for: Straight Jump (Section 7.1/7.2), Running Jump (Section 7.3/7.4).

Play **forward** on ascent (frames 1 → N), then **reverse** on descent (frames N → 1). This eliminates the need for a separate set of falling frames and creates a natural gravity arc.

```
if (velocity_y < 0) : //ascendingplay_forward(frames)if(velocity_y > 0) :
//descendingplay_reverse(frames)if(velocity_y == 0) : //apexhold(last_frame)
```

14.3 1

Technique: Modular Animation

(*Dragon's Crown*, *Metal Slug X*)

The slow-walk and fast-run lower bodies (Sections 3.2 and 4) are **shared across ALL weapon configurations**. This avoids duplicating lower-body frames for every weapon variant.

⚡ CRITICAL

- When the weapon changes, **only swap the upper body**.
- The lower body stays the same.
- **Exception 1:** Crouching has its own lower body (Section 8.3).
- **Exception 2:** Injured walk has its own lower body (defined in future sections).

14.4 1

Technique: Squash & Stretch Flash

(*Metal Slug*, *Contra*)

For all fire animations the first three frames represent the muzzle flash / recoil burst: **dark flash** → **blurred flash** → **escape**.

After frame 3 the character transitions back to the adjusted / held fire pose.

⚡ CRITICAL

- **Pistol:** USE ALL FRAMES (do not loop only first 3 for auto-fire).
- **Machine gun:** 4-frame burst loop (covered in the next document).
- **All other firing:** all frames play once, then hold the last frame.

14.5 1

Technique: Procedural Tilt Blending

(*Broforce, Terraria*)

Swimming upper body has 3 tilt states: **Straight** (0°), **Slight Tilt**, **More Tilt**. Each tilt state has 3 sub-states: Still, Active Swim, Swim Up.

```
if (velocityy < 0) : playforward//goingup if (velocityy > 0) : playreverse//goingdown
```

Match body tilt to gun angle for realism:

- Straight tilt → gun aimed horizontally
- Slight tilt → gun aimed slightly up / down
- More tilt → gun aimed steeply up / down

14.6 1

Technique: Sprite Layering / Z-Order Compositing

(*Castlevania, zombie-type characters*)

The zombie hand is a **separate sprite drawn on top of the body**. Two hand types: **Straight Hand** and **Up Hand**. Each has its own frames for: Simple Movement, Fast Movement, Jump, and Fire.

🎮 Rule

Match hand frame index **exactly** to body frame index (1:1 synchronisation).

14.7 1

Load the mummy ready phase **every time** the character is not walking. This must be checked each frame.

```
if (iswalking == false) : playreadyphaseloop()
```

14.8 1

When crouching, the lower body is always attached **directly below** the crouching upper body (see Section 8). Crouching fire, sliding fire, and crouch melee all have their lower body attached in this manner.

📌 Note

Do not substitute the running lower body for the crouch lower body — they have different heights and hip positions, and mixing them causes a visible seam jump.

14.9 1

When the biome changes from any biome to the **Aquatic biome**, the character plays the **entering-water jump animation** before the swim loop activates.

```
if (biome_changed_to == AQUATIC) : play_once(enter_water_jump_anim) then_start(swim_loop)
```

14.10 1

Rule	Description
Hip Anchor Constant	The Y offset between upper and lower body never varies within a weapon config.
One Weapon = One Top Body	Swapping weapons only replaces the top body layer.
Loop vs One-Shot	Idle, run, walk, and crouch-walk loop. All others are one-shot unless stated.
Palindrome Jumps	All jump animations play forward on rise, reverse on fall.
Pistol: All Frames	Never truncate the pistol fire sequence to 3-frame loop.
Frame-Sync Layers	Top and bottom body frame indices must always match.
Crouch Has Own Lower	Never reuse the run/walk lower body for crouch states.
Biome Entry Plays Once	Water entry animation fires exactly once per biome transition.